

Chapter 8.

$$8.1. \quad UV^t = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_n v_1 & u_n v_2 & \dots & u_n v_n \end{bmatrix} \begin{matrix} \leftarrow u_1 V \\ \leftarrow u_2 V \\ \vdots \\ \leftarrow u_n V \end{matrix}$$

$$8.2. \quad \begin{bmatrix} 1 & 0 & 4 \\ 2 & 0 & 8 \\ -1 & 0 & -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 4 \end{bmatrix}$$

$$8.3. \quad \text{No. Since } \begin{bmatrix} 3 \\ 6 \\ 8 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ for any } \lambda.$$

$$8.4. \quad A = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix} - \begin{bmatrix} -1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & -1 \end{bmatrix} = I_{5 \times 5} - \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$\begin{matrix} \leftarrow u \\ \leftarrow v^t \end{matrix}$

$$8.5. \quad By = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 3 \\ 3 \end{bmatrix} \Rightarrow y = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 3 \\ 3 \end{bmatrix} \text{ and } Bz = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} \Rightarrow z = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} \text{ Since } B = I_{5 \times 5}$$

$$x = y + \left(\frac{v^t y}{1 - v^t z} \right) z = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 3 \\ 3 \end{bmatrix} + \left(\frac{6}{1 - (-2)} \right) \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \\ -2 \\ -2 \\ -2 \end{bmatrix} \\ = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

$$8.6. \quad A = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 3 & -1 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 3 & -1 & 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

$\begin{matrix} \leftarrow B & \leftarrow u & \leftarrow v^t \end{matrix}$

Then using Sherman-Morrison algorithm, we can find.

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$8.7. \quad X = \begin{bmatrix} 25 \\ 28 \\ 27 \\ 20 \\ 5 \end{bmatrix}$$

$$8.8. \quad \begin{bmatrix} 1 & 2 & -1 & -1 \\ -1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & -1 \\ -1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \xleftarrow{B} - \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= B - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \xleftarrow{u} [0 \ 0 \ 1 \ 0] \xleftarrow{v^t}$$

$$By = \begin{bmatrix} -4 \\ 2 \\ 4 \\ 3 \end{bmatrix} \Rightarrow y = \begin{bmatrix} 7 \\ -5 \\ 4 \\ 1 \end{bmatrix} \text{ using L-U factorization of } B$$

$$Bz = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow z = \begin{bmatrix} -3 \\ 2 \\ -1 \\ 0 \end{bmatrix}$$

$$X = y + \left(\frac{v^t y}{1 - v^t z} \right) z = \begin{bmatrix} 7 \\ -5 \\ 4 \\ 1 \end{bmatrix} + \left(\frac{24}{1 - (-1)} \right) \begin{bmatrix} -3 \\ 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \\ 4 \\ 1 \end{bmatrix} + \begin{bmatrix} -6 \\ 4 \\ -2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

Chapter 9.

9.1. $\|X\|_1 = 18$

$$\|X\|_2 = \sqrt{86}$$

$$\|X\|_\infty = 6.$$

9.2. Proof: $\|X\|_2 = \sqrt{X_1^2 + X_2^2 + \dots + X_n^2} \geq \sqrt{\max_{1 \leq i \leq n} X_i^2} = \sqrt{\|X\|_\infty^2} = \|X\|_\infty$

Also,

$$\|X\|_2 = \sqrt{X_1^2 + X_2^2 + \dots + X_n^2} \leq \sqrt{n \cdot \max_{1 \leq i \leq n} X_i^2} = \sqrt{n} \sqrt{\max_{1 \leq i \leq n} X_i^2} = \sqrt{n} \|X\|_\infty$$

9.3. Proof: $\|\alpha A\| = \max_{x \neq 0} \frac{\|\alpha A x\|}{\|x\|} = \max_{x \neq 0} \frac{\|\alpha(Ax)\|}{\|x\|}$

$$= |\alpha| \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = |\alpha| \|A\|$$

9.4. Proof: $\|I\| = \max_{x \neq 0} \frac{\|Ix\|}{\|x\|} = \max_{x \neq 0} 1 = 1$

9.5. $\|A\|_1 = 17, \quad \|A\|_\infty = 18.$

9.6. Proof: $X = (X - Y) + Y$

According to the triangle inequality, we have.

$$\|X\| \leq \|X - Y\| + \|Y\|$$

$$\Rightarrow \|X\| - \|Y\| \leq \|X - Y\|. \quad - \textcircled{1}$$

On the other hand, by symmetry, we also can get.

$$\|Y\| - \|X\| \leq \|X - Y\| \quad - \textcircled{2}$$

Combining $\textcircled{1}$ & $\textcircled{2}$, we obtain.

$$|\|X\| - \|Y\|| \leq \|X - Y\|$$

$$9.7. \quad \|A\|_1 = \|A^T\|_\infty$$

$$\begin{aligned}
 9.8. \quad \|uv^T\|_2 &= \max_{x \neq 0} \frac{\|uv^T x\|_2}{\|x\|_2} = \max_{x \neq 0} \frac{\|(v^T x)u\|_2}{\|x\|_2} \\
 &= \max_{x \neq 0} \frac{|v^T x| \|u\|_2}{\|x\|_2} \quad (|v^T x| \leq \|v\|_2 \|x\|_2) \\
 &\leq \max_{x \neq 0} \frac{\|v\|_2 \|x\|_2 \|u\|_2}{\|x\|_2} \quad \text{Cauchy-Schwarz inequality} \\
 &= \|v\|_2 \|u\|_2
 \end{aligned}$$

$$9.9. \quad \|A\|_1 = \|A_2\| = \|A\|_\infty = \max_{1 \leq i \leq n} |d_i|$$

$$9.10. \quad (i) \text{ Proof: } \|X\|_2^2 = \sum_{i=1}^n x_i^2 = [x_1 \ x_2 \ \dots \ x_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = X^T X$$

$$\begin{aligned}
 (ii) \text{ Proof: } \|X+Y\|_2^2 &= (X+Y)^T (X+Y) \\
 &= (X^T + Y^T)(X+Y) \\
 &= X^T X + X^T Y + Y^T X + Y^T Y \\
 &= \|X\|_2^2 + X^T Y + (X^T Y)^T + \|Y\|_2^2
 \end{aligned}$$

Since $X^T Y = 0$, we get.

$$\|X+Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2$$

$$9.11. \text{ Proof. } \|A\|_1 = \max_{x \neq 0} \frac{\|AX\|_1}{\|x\|_1}$$

$$\text{Since } \|AX\|_1 \leq \sqrt{n} \|AX\|_2,$$

$$\|x\|_1 \geq \|x\|_2,$$

we get

$$\|A\|_1 \leq \max_{x \neq 0} \frac{\sqrt{n} \|AX\|_2}{\|x\|_2} = \sqrt{n} \max_{x \neq 0} \frac{\|AX\|_2}{\|x\|_2} = \sqrt{n} \|A\|_2. \quad \text{--- ①}$$

On the other hand, since

$$\|AX\|_1 \geq \|AX\|_2, \quad \|X\|_1 \leq \sqrt{n} \|x\|_2,$$

we get again

$$\|A\|_1 \geq \max_{x \neq 0} \frac{\|AX\|_2}{\sqrt{n} \|x\|_2} = \frac{1}{\sqrt{n}} \|A\|_2. \quad \text{--- ②}$$

Combining ① & ③, we get.

$$\frac{1}{\sqrt{n}} \|A\|_2 \leq \|A\|_1 \leq \sqrt{n} \|A\|_2$$

9.12. Proof: Similar to the Problem 9.11.

9.13. Proof: Similar to the Problem 9.11.

Chapter 10.

10.1. $k_\infty(A) = 21$.

10.2. (a) $k_1(A) = 54$ $k_\infty(A) = 54$.

(b) $k_1(A) = 20$ $k_\infty(A) = 20$

(c) $k_1(A) = 187$ $k_\infty(A) = 144$.

10.3. Proof: $k(A) = \|A\| \|A^{-1}\| = \|A^{-1}\|^{-1} \|A^{-1}\| = k(A^{-1})$

10.4. Proof: $k_1(A) = \|A\|, \|A^{-1}\|,$
 $= \|A^t\|_\infty \| (A^{-1})^t \|_\infty$
 $= \|A^t\|_\infty \| (A^t)^{-1} \|_\infty$
 $= k_\infty(A^t)$

10.5. $\frac{1}{n^2} k_\infty(A) \leq k_1(A) \leq n^2 k_\infty(A)$

10.6. $k_1(A) = k_2(A) = k_\infty(A) = \frac{\max_{1 \leq i \leq n} |d_i|}{\min_{1 \leq i \leq n} |d_i|}$

10.7. $X - X' = \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} -1.3 \\ 1.2 \end{bmatrix} = \begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix}$
 $b - b' = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1.1 \\ 0.9 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$

For ∞ -norm,

$$\|x - x'\|_{\infty} = 0.3, \quad \|x\|_{\infty} = 1$$

$$\|b - b'\|_{\infty} = 0.1, \quad \|b\|_{\infty} = 1$$

$$K_{\infty}(A) = 2.1$$

$$\frac{\|x - x'\|_{\infty}}{\|x\|_{\infty}} = 0.3 \quad \frac{\|b - b'\|_{\infty}}{\|b\|_{\infty}} = 0.1$$

$$0.3 < 2.1(0.1) = 2.1.$$

10.8. Similar to 10.7.

$$\begin{aligned} 10.9. \quad r = b - Ax' &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.999 & 1 \\ 1 & 0.999 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1.998 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.002 \\ 0 \end{bmatrix} \end{aligned}$$

The residual equation $Ae = r$ is

$$\begin{bmatrix} 0.999 & 1 \\ 1 & 0.999 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 0.002 \\ 0 \end{bmatrix}$$

10.10. The exact solution to the equation in 10.9. is

$$x = \begin{bmatrix} \frac{2000}{1999} \\ \frac{2000}{1999} \end{bmatrix} \Rightarrow x - x' = \begin{bmatrix} -\frac{1998}{1999} \\ \frac{2000}{1999} \end{bmatrix}$$

$$\frac{\|x - x'\|_{\infty}}{\|x\|_{\infty}} = \frac{\frac{2000}{1999}}{\frac{2000}{1999}} = 1$$

$$\frac{\|b - b'\|_{\infty}}{\|b\|_{\infty}} = \frac{0.002}{2} = 0.001.$$

$$10.11. \quad \|A\|_{\infty} = 1999.$$

$$1 < 1999(0.001) = 1.999.$$

$$10.12. \quad \frac{355}{113} \approx 3.14159292 \Rightarrow 0.00000005 \times 10^0 \leq \left| \pi - \frac{355}{113} \right| \leq 0.0000005 \times 10^0.$$

so the approximation has 7 significant digits.

$$10.13 \quad 0.0001352 = 1.352 \times 10^{-4}. \text{ The approximation has 2 significant digits.}$$